

Dear Students,

As we conclude our exploration of how macro-phenomena emerge from micro-interactions, it is time to apply these concepts. For your final Course Project, you will become the architect of an artificial society.

Please read the following instructions carefully, as this project constitutes a significant portion of your grade.

1. Paper Selection (Action Required)

I have shared a folder containing a curated set of foundational papers (covering Opinion Dynamics, Conflict, Hierarchy, etc.).

- **Selection Process:** Selection is on a **First-Come-First-Serve basis**.
- **How to Claim:** Once you have decided on a paper, post the title immediately in the class **WhatsApp group**. Once a paper is claimed by a student, it is locked and cannot be chosen by anyone else.
- **The "Outside Paper" Rule:** You are strongly advised to choose from the uploaded list. In *rare* cases, you may propose a paper outside this list, but you must provide a strong justification for why it is relevant and feasible. This requires my explicit approval.

2. The "Simplicity" Trap

Some of the papers in the folder may appear simple, and you may even find existing NetLogo or Python code for them online. **Do not let this mislead you.**

- **Evaluation Criteria:** Your grade is **not** based on merely running existing code. It is based on **how much you add** to the existing work and the depth of your understanding.
- The original paper is only the *starting point*. Whether the base model is complex or simple, the rigorousness of your **extension** (The "What If?" scenario) and your analysis of the results will determine your score.

3. The Methodology

- **Phase 1: Replication.** Reproduce the main results/graphs of the selected paper. If the paper says polarization occurs at a specific threshold, your code must demonstrate this.
- **Phase 2: Extension.** Introduce a novel variable, rule, or constraint. (e.g., *What happens if we add "stubborn" agents? What if the network structure changes?*)
- **Phase 3: Analysis.** Run systematic experiments (averages over multiple runs) to compare your Extension against the Baseline.

4. Final Deliverable: The Conference Paper

Instead of a standard lab report, every student must write a **Conference Paper**.

- **Format:** The specific template/format will be shared with you later.
- **Academic Integrity:** Your paper will be strictly checked against **Plagiarism and AI-generation tools**. You are expected to write in your own words, demonstrating your own analytical thinking. [Don't check this instruction document against AI-generation. It is 95% AI-generated with my inputs ;-)]

5. Weightage Warning

Please remember that this project carries a **weightage of 40%** of your total course grade. It requires significant thought, coding effort, and analysis. Do not leave this until the last minute.

Next Steps:

1. Access the shared folder.
2. Read the abstracts.
3. Claim your paper on WhatsApp.

Happy Modeling.

- Harsha